1 - AURA

Good morning, everyone. We're Group 5, and we're excited to introduce AURA, our AI-powered jewelry visual search engine. AURA lets you search with your eyes, not your words, transforming how we discover jewelry. This project represents our deep dive into artificial intelligence and deep learning applications. We believe AURA offers a significant leap forward in visual search technology.

2 - The Problem with Traditional Search

Building on that, let's consider the core issue AURA addresses: the semantic gap in traditional jewelry search. Users often struggle with specialized vocabulary, leading to frustration and missed discoveries. Terms like 'filigree' or 'pavé' are rarely known by the average consumer. This creates a mismatch between how people describe what they want and how products are tagged. Visual complexity makes it even harder; you can't easily describe intricate details with text. This inefficiency often results in 'zero results' for visually specific searches. Now, let's see how AURA solves this.

3 - Introducing AURA

So, how does AURA tackle these challenges? We introduce AURA: Purely Visual Intelligence. See it, search it, find it. Users simply upload a photo, and our system instantly extracts a 'Visual Fingerprint' from that image. This eliminates the need for manual tagging or complex text queries entirely. AURA then instantly finds the top 10 most similar designs from thousands of options. This direct visual matching bypasses all the problems of keyword-based search. Next, we'll explore the technical backbone that makes this possible.

4 - System Architecture

To achieve this, AURA employs a robust dual-phase processing pipeline. Our offline phase pre-processes the entire dataset, extracting visual features using EfficientNetB0 and building a FAISS index. This creates a 'Digital Warehouse' of visual fingerprints. Then, in the online phase, our FastAPI backend handles real-time query uploads. It quickly matches the new visual fingerprint against our pre-built index. This seamless data flow ensures rapid and accurate results. Let's now look at the dataset that powers AURA.

5 - Dataset — RingFIR

Our system is built upon a robust foundation: the RingFIR dataset. This collection features 2,605 earring photographs across 46 distinct design categories. All images are standardized to 224x224 pixels for consistent model input. Crucially, AURA requires zero manual tagging; the system learns purely from these visual patterns. This diversity, from simple studs to complex drop earrings, ensures a comprehensive search experience. This dataset is key to AURA's ability to learn and recognize intricate visual details.

6 - Feature Extraction — EfficientNetB0

Now, let's dive into how AURA actually sees. We call this the Visual Fingerprint, powered by EfficientNetB0. This AI acts as our system's eye, translating complex pixel data into a concise, meaningful representation. It takes each image and converts it into a list of 1,280 descriptive numbers. This process captures the unique visual essence of each piece, things like curvature, texture, and even gem settings. We leverage transfer learning, specifically Global Average Pooling, to extract these compact embeddings without needing to retrain the entire model. This gives us state-of-the-art accuracy with minimal computational overhead. With these visual fingerprints created, we're ready to store them efficiently.

7 - Similarity Search — FAISS IndexFlatL2

Once we have these visual fingerprints, we need a way to store and search them quickly. This brings us to the Digital Warehouse, which uses FAISS IndexFlatL2. FAISS maps all 2,605 visual fingerprints into a multi-dimensional space. Think of it as plotting each piece of jewelry on a giant mathematical map. We use IndexFlatL2 for exhaustive search, ensuring perfect accuracy in finding the nearest neighbors. This means we're measuring the precise mathematical distance between fingerprints. A smaller distance means higher visual similarity. We then convert this distance into an intuitive percentage score, making it easy to understand. This system provides a super-fast digital filing cabinet for near-instant retrieval. Now, let's look at the underlying technologies that make all this possible.

8 - Technology Stack

Our technology stack is built for high performance and scalability. For the core intelligence, we use TensorFlow and Keras. These frameworks load our pre-trained EfficientNetB0 model, enabling high-dimensional feature extraction. For search and backend performance, we rely on FastAPI for asynchronous REST services. And of course, Meta AI's

FAISS provides sub-millisecond vector similarity search. On the user experience side, we've opted for vanilla HTML5, CSS3, and JavaScript. This creates a lightweight, responsive interface. Finally, Docker ensures consistent, containerized deployment across any environment. This robust combination allows us to deliver a seamless and efficient user experience. And speaking of user experience, let's explore the premium design of our interface.

9 - User Interface — Premium Experience

The user interface for AURA offers a truly premium experience. We've designed it with a dark luxury aesthetic, featuring an obsidian background and champagne gold accents. This creates a sophisticated, high-end feel, perfectly suited for jewelry discovery. We've also incorporated glassmorphism, using semi-transparent frosted-glass containers for a modern touch. Key features include a drag-and-drop upload zone with an instant local image preview. This is powered by the FileReader API. The entire design is fully responsive, with animated result cards and detailed modals. This ensures a zero-friction user flow, from image upload to dynamic results. It's all built with vanilla JavaScript for maximum performance. Next, we'll examine the project's structure and execution.

10 - Project Structure & Pipeline

Now, let's look at how we structured this project to ensure a smooth and efficient workflow. Our development pipeline is clearly organized, allowing for logical progression from data ingestion to deployment. We have distinct scripts for generating embeddings and building the FAISS index, which are critical offline steps. And our FastAPI server, driven by main.py, handles all the real-time search requests. This modular approach ensures maintainability and scalability, which are key for any robust AI system. This structured approach underpins the impressive performance we're about to discuss.

11 - Results & Conclusion

So, what did we achieve with this carefully constructed pipeline? We've delivered real-time performance, with query responses under 50 milliseconds, which is crucial for a responsive user experience. Importantly, our system operates with zero manual tagging, fully automating the process from raw image to searchable vector. This means we've successfully built a system that truly understands visual patterns. Looking ahead, we plan to expand AURA to include other jewelry categories and scale our indexing for millions of items. We also envision a future where we combine visual search with natural language queries, creating an even more powerful tool. This leads us to our final thoughts on AURA.

12 - Thank You

In conclusion, AURA represents a significant step forward in AI-powered visual search for jewelry. We've shown that you can truly search with your eyes, not your words, overcoming the limitations of traditional text-based systems. We believe this project demonstrates the power of AI to transform how we interact with complex visual data. Thank you for your time and attention. We're now open for any questions you may have.